myob PayGlobal

Previous Next Print

Logging

Multiple tenants can be run within a single workflow engine. There are various configurations that one can use for logging a tenant, driven by the Workflow *log4net.config* configuration file, found in the Workflow application folder. The most common logging scenarios, with instructions as to how to set these up, are explained below.

Multiple workflow tenants logging to a single log file

The *log4net.config* file by default will log to a file called *log.txt* located in the Logs subfolder of the workflow application folder. This location is controlled by the appender entry in the *log4net.config* file as below

```
<appender name="FileAppender" type="log4net.Appender.RollingFileAppender">
  <file value="Logs/log.txt" />
  <preserveLogFileNameExtension value="true" />
  <appendToFile value="true" />
  <rollingStyle value="Composite" />
  <datePattern value="yyyy-MM" />
  <maxSizeRollBackups value="12" />
  <maximumFileSize value="10MB" />
  <staticLogFileName value="true" />
  <lockingModel type="log4net.Appender.FileAppender+MinimalLock" />

  <layout type="PayGlobal.Shared.Logging.Log4Net.Layout.PayGlobalPatternLayout, PayGlobal.Shared.Logging">
    <conversionPattern value="%date{yyyy-MM-ddTHH:mm:ss.fffzzz} | %-5level | %-36.36tenantId | %-36.36activityId | %logger | %message%newline" />
  </layout>

</appender>
```

To configure multiple tenants to point to this one appender file, each tenant should have a *<log4NetConfigFile>* line in its own settings section within the *tenants.config* file (which contains setup information for each tenant), as below. This line indicates that the tenant will use the settings in the *log4net.config* file to log messages.

```
<tenants>

  <tenant id="ba81e050-ec65-11df-98cf-0800200c9a66">
    <settings>
      <log4NetConfigFile>log4net.config</log4NetConfigFile>
```

In order to identify which log message line belongs to a specific tenant, the *Tenant Id* Guid, as defined in the *tenants.config file*, is written to the log as in the sample highlighted *log.txt* line below (the grey highlighted text is the *Tenant Id* for that log line.

```
95  2023-07-20T07:46:26.043+12:00 | INFO  | ba81e050-ec65-11df-98cf-080020111111 | cd5d4324-a13e-4d53-9e0a-7809a671ad22 | PayGlobal.Workflow.Cryptography.TenantConfigurationMar
96  2023-07-20T07:46:26.043+12:00 | INFO  | ba81e050-ec65-11df-98cf-080020111111 | cd5d4324-a13e-4d53-9e0a-7809a671ad22 | PayGlobal.Workflow.Core.Application.Configuration.Tena
97  2023-07-20T07:46:26.044+12:00 | INFO  | ba81e050-ec65-11df-98cf-080020111111 | cd5d4324-a13e-4d53-9e0a-7809a671ad22 | PayGlobal.Workflow.ProcessFlow.Service.ExpiredBookmar
98  2023-07-20T07:46:26.044+12:00 | INFO  | ba81e050-ec65-11df-98cf-080020111111 | cd5d4324-a13e-4d53-9e0a-7809a671ad22 | PayGlobal.Workflow.ProcessFlow.Service.ExpiredBookmar
99  2023-07-20T07:46:26.045+12:00 | INFO  | ba81e050-ec65-11df-98cf-080020111111 | cd5d4324-a13e-4d53-9e0a-7809a671ad22 | PayGlobal.Workflow.Wcf.Infrastructure.ConcurrentServic
100 2023-07-20T07:46:26.047+12:00 | INFO  | ba81e050-ec65-11df-98cf-080020111111 | cd5d4324-a13e-4d53-9e0a-7809a671ad22 | PayGlobal.Workflow.GeneralWorkflow.Service.Notificatic
101 2023-07-20T07:46:26.048+12:00 | INFO  | ba81e050-ec65-11df-98cf-080020111111 | cd5d4324-a13e-4d53-9e0a-7809a671ad22 | PayGlobal.Workflow.GeneralWorkflow.Service.Notificatic
102 2023-07-20T07:46:26.048+12:00 | INFO  | ba81e050-ec65-11df-98cf-080020111111 | cd5d4324-a13e-4d53-9e0a-7809a671ad22 | PayGlobal.Workflow.Wcf.Infrastructure.ConcurrentServic
```

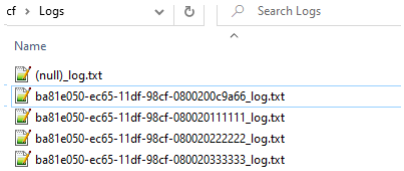
Multiple workflow tenants each logging to a separate log file

There are a couple of ways to achieve this, an automated or a manual way.

The automated method involves editing appender section in the *log4net.config file*, replacing the *<file value="Logs/log.txt" />* line with *<file type="log4net.Util.PatternString" value="Logs/%property{TenantIdentifier}_log.txt" />*, as in the example below (note that the *<file value="Logs/log.txt" />* line has simply been commented out rather than deleted).

```
<appender name="FileAppender" type="log4net.Appender.RollingFileAppender">
  <file type="log4net.Util.PatternString" value="Logs/%property{TenantIdentifier}_log.txt" />
  <!--<file value="Logs/log.txt" />-->
  <preserveLogFileNameExtension value="true" />
  <appendToFile value="true" />
  <rollingStyle value="Composite" />
  <datePattern value="yyyy-MM" />
  <maxSizeRollBackups value="12" />
  <maximumFileSize value="10MB" />
  <staticLogFileName value="true" />
  <lockingModel type="log4net.Appender.FileAppender+MinimalLock" />
```

The following log splits are then effected in the Log output folder, one file for each tenant entered in the *tenants.config file*.



Note that an extra *(null)_log.txt file* is created. There are occasions when the system cannot log to a tenant specific log file. For example, errors generated when tenants are created or loading. In this situation there may be no tenant log file to write to or the tenant may not be in a state where it can log. This *(null)_log.txt* file is a repository for log lines that fulfill these criteria.

A more manual process to split each tenant into its own log file is to create a *log4net.config file* for each tenant, with a different appender file name, and then reference this *log4net.config* file in each tenant section in the tenant.config. This is more time consuming and complicated and prone to error if global changes need to be made to config file files.

Multiple workflow tenants logging to a single log file with one or more tenants requiring temporary DEBUG log level

By default, logging is carried out at the INFO log4net log level, which gives a good balance between writing messages of use and importance, versus quantity of log messages written.

At times it may be useful to capture more logging information in order to track down problems that may be occurring. If only a single log file is used for multiple tenants, one does not wish to place all tenants in the DEBUG state, as this will lead to a large number of unnecessary log messages being written for all tenants rather than the one or two tenants under scrutiny. The following method is thus recommended.

1. Copy the existing *log4net.config* file to the same folder as the original and name it, say, *log4net_DEBUG.config*.
2. In this new config file, change the log levels in the appropriate *<root>* and *<logger>* sections to the desired log level (often DEBUG).
3. It is also recommended that the appender file name in the new DEBUG config file be changed to a separate file to avoid log file write contention.
4. In the *tenants.config* file, reference the newly created *log4net_DEBUG.config* file in the *<log4netConfigFile>* section for the appropriate tenant(s).
5. The tenants thus changed should now log at the DEBUG level, preferably into a new log file, while the other tenants should continue to log to the original log file at the INFO level.

Deprecated configuration (for existing installations)

Some existing clients may be using a deprecated *<logFilePath>* logging setting, for example *<logFilePath>C:\Logs\Workflow.4.0\</logFilePath>*. If this setting is still present and active in any tenant in the *tenants.config* file, a warning message is written and the following will occur for logging.

1. The *<logFilePath>* entry will be ignored
2. Logging will be configured according to the *log4net.config* file in the workflow application folder
3. The warning message will be written to the log file generated due to the above configuration

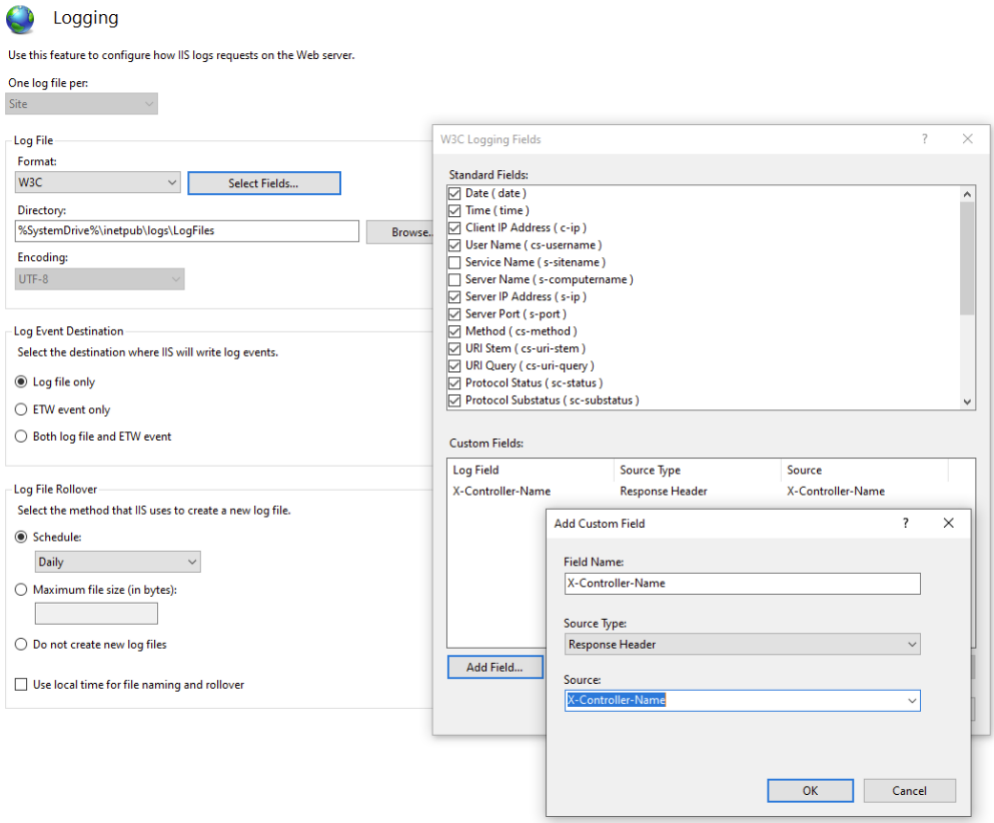
Performance Logging

Additional logging has been added to measure the performance of some features so they can be monitored for future improvements.

To enable this feature, the following tag will need to be added to each tenant in the Tenants.config file and then IIS may need to be reset:

<enablePerformanceLogging>True</enablePerformanceLogging>

In addition to this, IIS logs have been added for some endpoints. To enable this, select "Logging" on the app pool then "Select Fields...". Add a new "Response Header" with the field name and source being "X-Controller-Name".


Logging
Use this feature to configure how IIS logs requests on the Web server.
One log file per:
Site
Log File
Format:
W3C
Select Fields...
Directory:
%SystemDrive%\inetpub\logs\LogFiles
Browse...
Encoding:
UTF-8
Log Event Destination
Select the destination where IIS will write log events.
Log file only
ETW event only
Both log file and ETW event
Log File Rollover
Select the method that IIS uses to create a new log file.
Schedule:
Daily
Maximum file size (in bytes):
Do not create new log files
Use local time for file naming and rollover
W3C Logging Fields
Standard Fields:
Date (date)
Time (time)
Client IP Address (c-ip)
User Name (cs-username)
Service Name (s-sitename)
Server Name (s-computername)
Server IP Address (s-ip)
Server Port (s-port)
Method (cs-method)
URI Stem (cs-uri-stem)
URI Query (cs-uri-query)
Protocol Status (sc-status)
Protocol Substatus (sc-substatus)
Custom Fields:
Log Field
Source Type
Source
X-Controller-Name
Response Header
X-Controller-Name
Add Field...
Add Custom Field
Field Name:
X-Controller-Name
Source Type:
Response Header
Source:
X-Controller-Name
OK
Cancel